SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 3 (TOPIC 3)

Theme: PERSONAL COMPUTER (PC) SECURITY CONTROLS, FIRMWARE, TPM & ACCESS MODELS

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 3.1 (Hard Disk & Power-On Passwords), Subtopic 3.2 (BIOS/UEFI & Secure Boot/TPM), Subtopic 3.3 (Login Passwords & Access Controls)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 6 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Show all intermediate formulas, binary conversions, architectural diagrams, command syntax, and arithmetic calculations clearly.
4. Where design questions require software configurations, provide exact command line syntax (Bash/DOS) or functional formulas (Excel) as specified.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL SECURITY ARCHITECTURE [25 MARKS]

Question 1 [5 Marks]: Differentiate between an ATA Storage Security Password (hard disk password) and an Operating System Login Password. If a physical hard disk is extracted from a stolen laptop and connected via a USB enclosure to an external forensic computer, explain precisely which password protects the data and why.

Question 2 [5 Marks]: Contrast the legacy BIOS (Basic Input/Output System) with the modern UEFI (Unified Extensible Firmware Interface) architecture. Focus on: (a) Execution mode (16-bit real mode vs 32/64-bit protected mode), (b) Partition table support (MBR vs GPT), and (c) Secure firmware verification mechanisms.

Question 3 [5 Marks]: Explain the role and architecture of the Trusted Platform Module (TPM 2.0). Define the Endorsement Key (EK), Storage Root Key (SRK), and Platform Configuration Registers (PCRs). What is 'Measured Boot'?

Question 4 [5 Marks]: Describe the three fundamental factors of authentication: Something you know, Something you have, and Something you are. Give two microcomputer examples of each factor. What is Multi-Factor Authentication (MFA), and why does it drastically reduce account takeover risk?

Question 5 [5 Marks]: Differentiate between Discretionary Access Control (DAC) and Mandatory Access Control (MAC). Illustrate how standard Linux POSIX permissions represent DAC, whereas SELinux enforces MAC.

SECTION B: QUANTITATIVE SECURITY & CRYPTOGRAPHIC ANALYSIS [35 MARKS]

Question 6 [10 Marks]: Password Entropy and Brute-Force Mechanics:
(a) Formulate the equation for password entropy H in bits: $H = L \cdot \log_2(N)$. [2 Marks]
(b) Calculate the entropy for Candidate Password A ('sua2026') consisting of 7 lowercase alphanumeric characters (N = 36). [3 Marks]
(c) Calculate the entropy for Candidate Password B ('Mkulima#Kwanza@2026!') consisting of 21 characters drawn from uppercase, lowercase, digits, and special symbols (N = 95). [3 Marks]
(d) Assuming an offline GPU cracking cluster capable of testing 1.0×10^{11} SHA-256 hashes per second, calculate the expected average cracking time (50% keyspace) for both passwords. [2 Marks]

Question 7 [10 Marks]: Whole Disk Encryption (WDE) Architecture:
Analyze Microsoft BitLocker and Linux LUKS (Linux Unified Key Setup).
(a) Explain the cryptographic hierarchy utilized in BitLocker: Full Volume Encryption Key (FVEK), Volume Master Key (VMK), and TPM sealing key. Why does the system avoid encrypting sector data directly with the user password? [5 Marks]
(b) Describe a 'Cold Boot Attack' against RAM. How does memory remanence allow physical attackers to extract encryption keys, and what hardware mitigations (e.g., TME - Total Memory Encryption) exist in modern CPUs? [5 Marks]

Question 8 [8 Marks]: UEFI Secure Boot Chain of Trust:
A new microcomputer workstation is powered on with Secure Boot enabled.
(a) Trace the complete cryptographic chain of trust from the hardware Root of Trust (Platform Key - PK) down to the operating system kernel execution. Define the roles of Key Exchange Key (KEK), db (Allowed Signatures Database), and dbx (Revoked Signatures Database). [5 Marks]
(b) Explain what happens when a user attempts to boot an unsigned Linux live distribution on a Secure Boot enabled system. How does the 'shim' bootloader signed by Microsoft 3rd Party UEFI CA solve this for major Linux distributions? [3 Marks]

Question 9 [7 Marks]: Data Sanitization Standards (NIST SP 800-88 Rev 1):
(a) Contrast the three levels of sanitization: Clear, Purge, and Destroy. [3 Marks]
(b) Why is a traditional 7-pass zero-fill overwrite (DoD 5220.22-M) completely ineffective and destructive when applied to Solid-State Drives (SSDs)? Explain how Cryptographic Erase (crypto-shredding) and ATA Enhanced Secure Erase solve this problem without degrading NAND cells. [4 Marks]

SECTION C: APPLIED ENTERPRISE SECURITY SCENARIOS [40 MARKS]

Question 10 [15 Marks]: Enterprise PC Security Hardening at SUA Administration:
The SUA Examination Directorate manages 30 microcomputers containing highly confidential university examination papers, grading sheets, and student transcripts.
(a) Design a comprehensive multi-layered security hardening policy covering: (i) Physical chassis lock and port blocking, (ii) BIOS/UEFI supervisor passwords and boot order locking, (iii) TPM-backed Full Disk Encryption with BitLocker/LUKS, and (iv) Operating system user account privileges. [8 Marks]
(b) Detail the configuration of Windows Group Policy Objects (GPO) or Linux PAM rules to enforce: 14-character minimum password length, account lockout after 5 failed attempts, automated screen lock after 5 minutes of inactivity, and disabling USB mass storage write access. [4 Marks]
(c) Outline an incident response protocol if one of the examination laptops is reported stolen from an academic office. [3 Marks]

Question 11 [15 Marks]: Linux Access Control & Sudoers Security Engineering:
A shared departmental research server at SUA DIIT hosts 40 student users and 4 research staff.
(a) Explain the permission triplet (Owner, Group, Others) and special permission bits (SetUID, SetGID, Sticky Bit). Give the numeric octal representation for a shared project directory where all group members can read/write files, but cannot delete each other's files. [6 Marks]
(b) Write the exact Linux commands to: (i) Create a group called `agric_research`, (ii) Add users `bertha` and `baraka`

to the group, (iii) Set ownership and permissions on `/opt/crop_data` to ensure group collaboration with sticky bit protection. [5 Marks]
(c) Explain why granting `ALL=(ALL:ALL) ALL` in `/etc/sudoers` violates the Principle of Least Privilege. Show how to restrict a user to running only `/usr/bin/systemctl restart apache2` without a password prompt. [4 Marks]

Question 12 [10 Marks]: Biometric Security Evaluation in Field Microcomputers:
SUA Agricultural Extension Officers utilize ruggedized field tablets equipped with fingerprint scanners and facial recognition cameras to register smallholder farmers across Morogoro.
(a) Define False Acceptance Rate (FAR) and False Rejection Rate (FRR). Explain the trade-off between FAR and FRR when calibrating biometric sensor sensitivity. [4 Marks]
(b) What is the Crossover Error Rate (CER) / Equal Error Rate (EER), and why is it used as the benchmark for biometric device comparison? [3 Marks]
(c) How can biometric authentication be vulnerable to 'presentation attacks' (spoofing with silicone fingers or photographs), and what hardware liveness detection techniques exist to mitigate this vulnerability? [3 Marks]